

# Canonical Project Context and Audit for the SJSU DATA298 Persona-Modeling Project

## Executive summary

The project should now be treated as an **experimental study of author/persona simulation using public author-generated text**, rather than as a collection of loosely related RAG and fine-tuning implementations. The central scientific question is not “Which architecture is coolest?” but:

**How much do increasingly structured forms of persona information—prompted profile, retrieved evidence, ontological structure, parameter adaptation, and combinations of these—improve a language model’s ability to reproduce a target author’s linguistic style, recurring positions, and evidence-grounded perspective without simply memorizing or reproducing promotional/noisy content?**

That framing fits the project direction described in this conversation and supports a clean sequence of controlled experiments. It also aligns with SJSU’s official DATA 298 assessment emphasis on complete data preparation, model comparison, model justification, literature/technology survey, end-to-end lifecycle completion, quantitative analysis, communication, and societal contribution. <sup>1</sup>

The recommended core model ladder is:

| Model     | Canonical experiment                | Main question                                                                                                                                |
|-----------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M0</b> | Base instruction model              | What does the unadapted model already know/do?                                                                                               |
| <b>M1</b> | Structured persona prompt           | How far can explicit profile/context prompting go without retrieval or training?                                                             |
| <b>M2</b> | Vector PersonaRAG                   | Does retrieving authentic author evidence improve persona fidelity?                                                                          |
| <b>M3</b> | Ontology-grounded PersonaRAG        | Does explicitly structured persona knowledge improve retrieval and perspective consistency over flat/vector RAG?                             |
| <b>M4</b> | QLoRA persona adaptation            | Can parameter-efficient training internalize style/perspective better than context-only methods?                                             |
| <b>M5</b> | QLoRA + ontology/persona RAG        | Are parametric persona learning and structured non-parametric memory complementary?                                                          |
| <b>M6</b> | Advanced retrieval-aware adaptation | Does RAFT-style training, optionally incorporating APR/CharLoRA concepts, improve robustness to relevant and distracting retrieved evidence? |

This ladder makes **Ontology-RAG a first-class controlled experiment at M3**, rather than an implementation detail or last-minute professor requirement. Ontology-grounded RAG is materially different from ordinary vector RAG: OG-RAG explicitly grounds retrieval in a domain ontology and structured relationships, while systems such as LightRAG and GraphRAG construct graph representations primarily from entities/relations in documents. <sup>2</sup> For this project, the ontology should represent **author, source text, topic, stance, style trait, entity, event, evidence, content type, and promotional status** rather than merely a generic entity graph.

**CharLoRA/APR is unusually relevant to this project**, because CharacterBot specifically attempts to capture both linguistic patterns and deeper author perspectives from an author corpus, using Authorial Perspective Reframing plus multitask persona adaptation. <sup>3</sup> However, reproducing CharacterBot as a mandatory core architecture would substantially increase data-generation and training complexity. It is better treated as an advanced M4/M6 branch after the simpler QLoRA experiment has established a reproducible baseline.

**Persona-Plug is also relevant but should be a comparator/extension, not a required core architecture.** It turns a user's historical texts into an input-aware user embedding while leaving the underlying LLM fixed. <sup>4</sup> That is attractive for low-cost personalization, but it answers a somewhat different question from author-style fine-tuning or ontology-guided retrieval.

The social-media cleaning problem should be solved through **labeling, weighting, quarantine, and provenance—not destructive deletion**. Promotional content sometimes genuinely reflects an author's voice; blindly deleting it could remove useful style signal, while blindly training on it could cause the model to produce calls-to-action, sponsor language, repeated templates, discount codes, or platform boilerplate. The correct design therefore preserves the raw corpus, assigns content-type labels and confidence scores, creates different inclusion policies for persona profiling, RAG, QLoRA, and evaluation, and permanently records why each item was accepted or quarantined.

There is one important limitation to the present audit. **This chat does not currently have direct read access to** `C:\Users\okevi\SJSU\DATA298\298P\docs\research`. The file-search system reports that no retrievable local files from that folder have been uploaded or connected. I therefore cannot truthfully claim to have enumerated or read every research paper, workbook, final report, or newly added slide deck in that Windows directory. The file-specific portions below are therefore a **canonical-context v0.9 scaffold**: primary literature and the official SJSU rubric are externally verified, while local-document-specific assertions are explicitly marked **UNVERIFIED** rather than invented. Once that directory is exposed to this conversation, the local-file rows are the pieces that should be promoted from v0.9 to canonical v1.0.

## Evidence base and research index

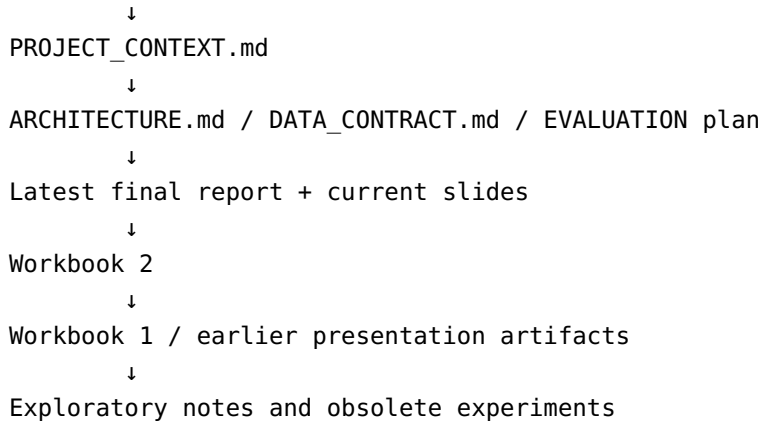
### Source-of-truth hierarchy

The repository should use the following precedence order:

Professor/rubric requirements

↓

Accepted DEC-xxx decisions



This ordering is a governance recommendation rather than a claim about the current contents of the local files. In particular, the latest report should **not silently override an accepted decision**; contradictions should generate a new `DEC-xxx` entry explaining what changed.

The official SJSU Applied Data Science assessment table supports making the data contract and experimental comparison first-class project artifacts: DATA 298A explicitly evaluates description of raw, preprocessed, transformed, and prepared datasets, visualization, comparisons among models, model justification, and literature/technology survey; DATA 298B evaluates the full project lifecycle, report/presentation/demo communication, quantitative problem solving, and project contribution to broader well-being. <sup>1</sup>

### RESEARCH\_INDEX table

The index below contains every **named local artifact presently known from this conversation**, plus the primary research sources needed to reconcile the named architectures. It does **not** claim that these are all files physically present in the directory.

| paper_id               | title / artifact | authors / year | source                 | topics                    | architecture relevance                  | cleaning / evaluation relevance           |
|------------------------|------------------|----------------|------------------------|---------------------------|-----------------------------------------|-------------------------------------------|
| <code>LOCAL-WB1</code> | Workbook 1       | UNVERIFIED     | Local project artifact | proposal, initial methods | Must be checked for original model list | Likely initial data/eval plan; unverified |
| <code>LOCAL-WB2</code> | Workbook 2       | UNVERIFIED     | Local project artifact | revised proposal/ method  | Must be compared with WB1               | Likely revised metrics/data plan          |

| paper_id     | title / artifact                                                 | authors / year        | source                     | topics                                | architecture relevance                                                           | cleaning / evaluation relevance                       |
|--------------|------------------------------------------------------------------|-----------------------|----------------------------|---------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------|
| LOCAL-DEMO   | Demo rubric                                                      | UNVERIFIED            | Local course artifact      | demonstration requirements            | Determines demo scope                                                            | May impose evaluation/display requirements            |
| LOCAL-PRES   | Presentation                                                     | UNVERIFIED            | Local project artifact     | architecture, results/proposal        | Known working context suggests multiple architecture variants; not file-verified | Check metrics/splits against report                   |
| LOCAL-REPORT | Final report                                                     | UNVERIFIED            | Newly added local artifact | current project narrative             | Should be highest-precedence project narrative after accepted decisions          | Check final methods/results claims                    |
| LOCAL-SLIDES | Final/current slides                                             | UNVERIFIED            | Newly added local artifact | presentation/demo                     | Check architecture names and diagrams                                            | Check results match report                            |
| RAG-2020     | Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks | Lewis et al., 2020    | Primary paper              | RAG, parametric/non-parametric memory | Establishes vanilla RAG baseline                                                 | Provenance and retrieval grounding                    |
| LORA-2021    | LoRA: Low-Rank Adaptation of Large Language Models               | Hu et al., 2021       | Primary paper              | PEFT, low-rank adaptation             | Foundation for QLoRA/CharLoRA                                                    | Efficient adaptation                                  |
| QLORA-2023   | QLoRA: Efficient Finetuning of Quantized LLMs                    | Dettmers et al., 2023 | Primary paper              | 4-bit PEFT, NF4                       | Practical persona fine-tuning baseline                                           | Paper stresses importance of high-quality tuning data |

| paper_id        | title / artifact                                                                      | authors / year                       | source        | topics                                        | architecture relevance                                  | cleaning / evaluation relevance                                   |
|-----------------|---------------------------------------------------------------------------------------|--------------------------------------|---------------|-----------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| LAMP-2023       | LaMP: When Large Language Models Meet Personalization                                 | Salemi et al., 2023                  | Primary paper | personalized LM evaluation, profile retrieval | Strong conceptual basis for M1/M2                       | Personalized generation benchmark structure                       |
| RAFT-2024       | RAFT: Adapting Language Model to Domain Specific RAG                                  | Zhang et al., 2024                   | Primary paper | retrieval-aware fine-tuning, distractors      | Best fit for optional M6                                | Tests robustness to imperfect retrieval                           |
| PERSONARAG-2024 | PersonaRAG: Enhancing Retrieval-Augmented Generation Systems with User-Centric Agents | Zerhoudi & Granitzer, 2024/2026 rev. | Primary paper | personalized RAG, user-centric agents         | Inspiration for author-aware retrieval/ranking          | Evaluates personalized retrieval behavior                         |
| PPLUG-2024      | LLMs + Persona-Plug = Personalized LLMs                                               | Liu et al., 2024/2026 rev.           | Primary paper | user embeddings, personalized generation      | Optional low-compute comparator                         | Tested on LaMP                                                    |
| OGRAG-2024      | OG-RAG: Ontology-Grounded Retrieval-Augmented Generation for Large Language Models    | Sharma, Kumar & Li, 2024             | Primary paper | ontology, hypergraph retrieval, attribution   | Primary source for required ontological experiment      | Supports explicit context attribution and fact-oriented retrieval |
| LIGHTRAG-2024   | LightRAG: Simple and Fast Retrieval-Augmented Generation                              | Guo et al., 2024/2025 rev.           | Primary paper | graph RAG, dual-level retrieval               | Potential implementation/comparator for graph retrieval | Retrieval accuracy/efficiency                                     |

| paper_id      | title / artifact                                                            | authors / year    | source                            | topics                                           | architecture relevance                          | cleaning / evaluation relevance                             |
|---------------|-----------------------------------------------------------------------------|-------------------|-----------------------------------|--------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------|
| GRAPHRAG-2024 | From Local to Global: A Graph RAG Approach to Query-Focused Summarization   | Edge et al., 2024 | Microsoft Research primary source | knowledge graph, communities, global queries     | Alternative graph-RAG paradigm                  | Useful for corpus-level thematic queries                    |
| CHARBOT-2025  | Beyond Profile: From Surface-Level Facts to Deep Persona Simulation in LLMs | Wang et al., 2025 | Primary paper                     | author simulation, APR, CharLoRA, style/ideology | Closest directly relevant advanced architecture | Uses content/style scoring, BLEU/ROUGE and human evaluation |

## Canonical YAML seed

The following should become `docs/RESEARCH_INDEX.yaml`. `enumeration_complete: false` is important until the physical Windows directory has actually been enumerated.

```
research_index:
  schema_version: "1.0"
  enumeration_complete: false
  local_root: "C:\\Users\\okevi\\SJSU\\DATA298\\298P\\docs\\research"
  note: >
    Local filenames and local-paper inventory have not yet been directly
    enumerated in this chat. Do not interpret this index as proof that all
    folder contents have been inspected.

  entries:
    - paper_id: "LOCAL-WB1"
      title: "Workbook 1"
      authors: null
      year: null
      source: "local project artifact"
      topics: ["proposal", "data", "architecture", "evaluation"]
      architecture_relevance: "Audit original architecture assumptions and
model list."
      data_cleaning_relevance: "Audit original preprocessing assumptions."
      evaluation_relevance: "Audit original splits and metrics."
      key_findings: "PENDING_LOCAL_INSPECTION"
      methods_we_can_reuse: []
      limitations: ["Content not accessible in current chat."]
      project_decisions_supported: []
```

```

citation: "Local course/project artifact"
local_file: "UNVERIFIED_FILENAME"

- paper_id: "LOCAL-WB2"
  title: "Workbook 2"
  authors: null
  year: null
  source: "local project artifact"
  topics: ["revised proposal", "architecture", "evaluation"]
  architecture_relevance: "Compare against Workbook 1 and later report."
  data_cleaning_relevance: "Check whether promotional/noisy content was
addressed."
  evaluation_relevance: "Check metric and split changes."
  key_findings: "PENDING_LOCAL_INSPECTION"
  methods_we_can_reuse: []
  limitations: ["Content not accessible in current chat."]
  project_decisions_supported: []
  citation: "Local course/project artifact"
  local_file: "UNVERIFIED_FILENAME"

- paper_id: "LOCAL-DEMO"
  title: "Demo rubric"
  authors: null
  year: null
  source: "local course artifact"
  topics: ["demo", "rubric", "acceptance criteria"]
  architecture_relevance: "May constrain which models must be demonstrated."
  data_cleaning_relevance: "May require visible pipeline evidence."
  evaluation_relevance: "Convert rubric items to demo acceptance tests."
  key_findings: "PENDING_LOCAL_INSPECTION"
  methods_we_can_reuse: []
  limitations: ["Content not accessible in current chat."]
  project_decisions_supported: []
  citation: "Local course artifact"
  local_file: "UNVERIFIED_FILENAME"

- paper_id: "LOCAL-PRES"
  title: "Project presentation"
  authors: null
  year: null
  source: "local project artifact"
  topics: ["architecture", "research questions", "results"]
  architecture_relevance: "Reconcile diagrams and model names with final
architecture."
  data_cleaning_relevance: "Check data-flow claims."
  evaluation_relevance: "Check reported metrics and experiment definitions."
  key_findings: "PENDING_LOCAL_INSPECTION"
  methods_we_can_reuse: []

```

```

limitations: ["Content not accessible in current chat."]
project_decisions_supported: []
citation: "Local project artifact"
local_file: "UNVERIFIED_FILENAME"

- paper_id: "LOCAL-REPORT"
  title: "Final report"
  authors: null
  year: null
  source: "local project artifact"
  topics: ["final narrative", "methods", "experiments", "results"]
  architecture_relevance: "Highest-priority local narrative once verified."
  data_cleaning_relevance: "Must accurately describe raw-to-prepared data
lifecycle."
  evaluation_relevance: "Must match actual experiment manifests and
results."
  key_findings: "PENDING_LOCAL_INSPECTION"
  methods_we_can_reuse: []
  limitations: ["Content not accessible in current chat."]
  project_decisions_supported: []
  citation: "Local final project report"
  local_file: "UNVERIFIED_FILENAME"

- paper_id: "LOCAL-SLIDES"
  title: "Final/current slides"
  authors: null
  year: null
  source: "local project artifact"
  topics: ["presentation", "architecture", "results"]
  architecture_relevance: "Must use same M0-M6 names as canonical
architecture."
  data_cleaning_relevance: "Must show data provenance and cleaning
succinctly."
  evaluation_relevance: "Numbers must trace to experiment artifacts."
  key_findings: "PENDING_LOCAL_INSPECTION"
  methods_we_can_reuse: []
  limitations: ["Content not accessible in current chat."]
  project_decisions_supported: []
  citation: "Local project slides"
  local_file: "UNVERIFIED_FILENAME"

- paper_id: "RAG-2020"
  title: "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks"
  authors:
    - Patrick Lewis
    - Ethan Perez
    - Aleksandra Piktus
    - Fabio Petroni

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- Vladimir Karpukhin
- Naman Goyal
- Heinrich Kuttler
- Mike Lewis
- Wen-tau Yih
- Tim Rocktaschel
- Sebastian Riedel
- Douwe Kiela

year: 2020

source: "arXiv:2005.11401"

topics: ["RAG", "retrieval", "non-parametric memory"]

architecture\_relevance: "Foundation for vector RAG baseline M2."

data\_cleaning\_relevance: "Retrieved corpus quality directly affects context."

evaluation\_relevance: "Supports retrieval/generation evaluation and provenance."

key\_findings: "Combines parametric generation with retrieved non-parametric memory."

methods\_we\_can\_reuse: ["top-k retrieval", "source-grounded generation"]

limitations: ["Not a persona-specific architecture."]

project\_decisions\_supported: ["DEC-006"]

citation: "Lewis et al. (2020), arXiv:2005.11401"

local\_file: "TO\_MATCH\_AFTER\_ENUMERATION"

- paper\_id: "QLORA-2023"
- title: "QLoRA: Efficient Finetuning of Quantized LLMs"
- authors: ["Tim Dettmers", "Artidoro Pagnoni", "Ari Holtzman", "Luke Zettlemoyer"]
- year: 2023
- source: "arXiv:2305.14314"
- topics: ["QLoRA", "PEFT", "4-bit training", "NF4"]
- architecture\_relevance: "Core fine-tuning baseline M4."
- data\_cleaning\_relevance: "High-quality tuning data is especially important."
- evaluation\_relevance: "Compare adapter model against context-only methods."
- key\_findings: "Backpropagates through a frozen quantized model into LoRA adapters."
- methods\_we\_can\_reuse: ["4-bit base", "LoRA adapters", "PEFT"]
- limitations: ["Does not itself define persona data construction."]
- project\_decisions\_supported: ["DEC-007"]
- citation: "Dettmers et al. (2023), arXiv:2305.14314"
- local\_file: "TO\_MATCH\_AFTER\_ENUMERATION"

- paper\_id: "LAMP-2023"
- title: "LaMP: When Large Language Models Meet Personalization"
- authors: ["Alireza Salemi", "Sheshera Mysore", "Michael Bendersky", "Hamed Zamani"]

```

year: 2023
source: "arXiv:2304.11406"
topics: ["personalization", "profile retrieval", "evaluation"]
architecture_relevance: "Supports profile/personalized retrieval
baselines."
data_cleaning_relevance: "User history/profile composition affects
personalization."
evaluation_relevance: "Useful conceptual benchmark for personalized
generation."
key_findings: "Profile augmentation improves personalized tasks in
reported baselines."
methods_we_can_reuse: ["retrieve relevant profile entries",
"personalization ablations"]
limitations: ["Tasks are not identical to author simulation."]
project_decisions_supported: ["DEC-006", "DEC-011"]
citation: "Salemi et al. (2023), arXiv:2304.11406"
local_file: "TO_MATCH_AFTER_ENUMERATION"

- paper_id: "PERSONARAG-2024"
title: "PersonaRAG: Enhancing Retrieval-Augmented Generation Systems with
User-Centric Agents"
authors: ["Saber Zerhoudi", "Michael Granitzer"]
year: 2024
source: "arXiv:2407.09394"
topics: ["PersonaRAG", "agents", "personalized retrieval"]
architecture_relevance: "Inspiration for author-aware retrieval M2/M3."
data_cleaning_relevance: "Persona/profile evidence quality affects
adaptation."
evaluation_relevance: "Compare personalized retrieval with non-
personalized RAG."
key_findings: "Introduces user-centric agents for adaptive personalized
retrieval."
methods_we_can_reuse: ["profile analysis", "contextual retrieval",
"document ranking"]
limitations:
- "Original system personalizes for users, not necessarily simulates an
author."
project_decisions_supported: ["DEC-006"]
citation: "Zerhoudi and Granitzer (2024), arXiv:2407.09394"
local_file: "TO_MATCH_AFTER_ENUMERATION"

- paper_id: "OGRAG-2024"
title: "OG-RAG: Ontology-Grounded Retrieval-Augmented Generation for
Large Language Models"
authors: ["Kartik Sharma", "Peeyush Kumar", "Yunqing Li"]
year: 2024
source: "arXiv:2412.15235"
topics: ["ontology", "hypergraph", "structured retrieval", "attribution"]

```

```

architecture_relevance: "Primary foundation for M3."
data_cleaning_relevance: "Ontology mapping requires reliable source
labels."
evaluation_relevance: "Supports fact/context attribution and structured
retrieval comparisons."
key_findings: "Grounds retrieved context using explicit domain ontology."
methods_we_can_reuse: ["ontology schema", "structured neighborhoods",
"attribution"]
limitations:
  - "Designed primarily for fact-oriented/domain workflows."
  - "Persona style and stance require adaptation of ontology concepts."
project_decisions_supported: ["DEC-008"]
citation: "Sharma, Kumar, and Li (2024), arXiv:2412.15235"
local_file: "TO_MATCH_AFTER_ENUMERATION"

- paper_id: "RAFT-2024"
title: "RAFT: Adapting Language Model to Domain Specific RAG"
authors:
  - Tianjun Zhang
  - Shishir G. Patil
  - Naman Jain
  - Sheng Shen
  - Matei Zaharia
  - Ion Stoica
  - Joseph E. Gonzalez
year: 2024
source: "arXiv:2403.10131"
topics: ["RAFT", "retrieval-aware fine-tuning", "distractors"]
architecture_relevance: "Optional advanced M6."
data_cleaning_relevance: "Requires explicit relevant and distractor
document construction."
evaluation_relevance: "Tests retrieval robustness rather than persona
fidelity alone."
key_findings: "Trains models to answer with useful documents while
handling distractors."
methods_we_can_reuse: ["gold document plus distractors", "retrieval-aware
SFT"]
limitations: ["Domain QA method, not intrinsically a persona model."]
project_decisions_supported: ["DEC-010"]
citation: "Zhang et al. (2024), arXiv:2403.10131"
local_file: "TO_MATCH_AFTER_ENUMERATION"

- paper_id: "PPLUG-2024"
title: "LLMs + Persona-Plug = Personalized LLMs"
authors:
  - Jiongnan Liu
  - Yutao Zhu
  - Shuting Wang

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- Xiaochi Wei
- Erxue Min
- Yu Lu
- Shuaiqiang Wang
- Dawei Yin
- Zhicheng Dou

year: 2024

source: "arXiv:2409.11901"

topics: ["personalization", "user embedding", "plug-in adaptation"]

architecture\_relevance: "Optional alternative to retrieval or per-author fine-tuning."

data\_cleaning\_relevance: "Historical content quality affects the aggregate embedding."

evaluation\_relevance: "Useful low-parameter personalization comparator."

key\_findings: "Aggregates user history into an input-aware personal embedding."

methods\_we\_can\_reuse: ["history encoder", "personal embedding"]

limitations: ["Additional embedder implementation; different research question from RAG."]

project\_decisions\_supported: ["DEC-012"]

citation: "Liu et al. (2024), arXiv:2409.11901"

local\_file: "TO\_MATCH\_AFTER\_ENUMERATION"

- paper\_id: "CHARBOT-2025"

title: "Beyond Profile: From Surface-Level Facts to Deep Persona Simulation in LLMs"

authors:

- Zixiao Wang
- Duzhen Zhang
- Ishita Agrawal
- Shen Gao
- Le Song
- Xiuying Chen

year: 2025

source: "arXiv:2502.12988"

topics: ["CharacterBot", "APR", "CharLoRA", "style", "ideology"]

architecture\_relevance: "Directly relevant advanced persona adaptation."

data\_cleaning\_relevance: "Requires curated author-authentic training material."

evaluation\_relevance: "Separates content/perspective and style evaluation."

key\_findings: "Uses APR plus multi-task CharLoRA for linguistic and perspective alignment."

methods\_we\_can\_reuse:

- "Authorial Perspective Reframing"
- "style-transfer pairs"
- "persona QA"
- "content/style evaluation"

```

limitations:
  - "Complex synthetic-data pipeline."
  - "Case study on a literary author differs from social-media authors."
project_decisions_supported: ["DEC-009", "DEC-010", "DEC-011"]
citation: "Wang et al. (2025), arXiv:2502.12988"
local_file: "TO_MATCH_AFTER_ENUMERATION"

```

The core metadata and method descriptions in those external entries come from the cited primary papers. RAG combines parametric and retrieved non-parametric memory; QLoRA trains low-rank adapters through a frozen 4-bit model; LaMP formalizes multiple personalization tasks; PersonaRAG adds user-oriented analysis/ranking agents; PPlug aggregates histories into personal embeddings; RAFT trains against useful and distracting retrieved documents; OG-RAG applies an explicit ontology; and CharacterBot combines APR and CharLoRA for author-persona modeling. <sup>14</sup>

## Artifact audit and canonical definitions

### What is verified versus pending

A complete local-document contradiction audit cannot be presented as completed until the actual files are readable. The authoritative comparison should therefore distinguish **verified evidence**, **conversation-level project facts**, and **pending document inspection**.

| Artifact / evidence | Canonical facts currently extractable                                | Conflict or outdated-assumption check                                                                | Unresolved decision                           | Proposed canonical treatment                                             |
|---------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------|
| Workbook 1          | Exists according to project context; contents presently inaccessible | Compare model list, dataset size, splits, metrics, stated research question with all later artifacts | Which assumptions have been superseded?       | Historical proposal; never override accepted later decisions             |
| Workbook 2          | Exists; contents inaccessible                                        | Compare directly with Workbook 1                                                                     | Whether changes were deliberate or accidental | Revised proposal layer                                                   |
| Demo rubric         | Exists locally; contents inaccessible                                | Compare with official program-level expectations                                                     | Exact demo requirements                       | Translate each rubric item into a reproducible demo acceptance criterion |

| Artifact / evidence      | Canonical facts currently extractable                                                                                                                                                                                               | Conflict or outdated-assumption check                                           | Unresolved decision                                               | Proposed canonical treatment                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Earlier presentation     | Exists; contents inaccessible                                                                                                                                                                                                       | Architecture names, model count, diagrams, result numbers                       | Which diagram is canonical?                                       | Presentation is explanatory, not source of technical truth                         |
| Final report             | Newly added according to user                                                                                                                                                                                                       | Check all architecture descriptions, data totals, splits, results and citations | Whether report contains final decisions or stale proposal text    | Highest-priority narrative, but must agree with decisions and experiment manifests |
| Current/final slides     | Newly added according to user                                                                                                                                                                                                       | Numbers and diagrams must match report/ repo                                    | Which experiments are actually complete?                          | Presentation projection of canonical context                                       |
| Official SJSU assessment | DATA 298A requires data-preparation reporting, visualization, model comparisons/justifications and technology survey; DATA 298B emphasizes full lifecycle, communication, quantitative work and societal contribution. <sup>1</sup> | Local rubrics may add specificity                                               | Exact local demo/report scoring                                   | Non-negotiable minimum alignment                                                   |
| Working project context  | Target-author/persona modeling; social-media data; promotional contamination concern; professor requested exploration of ontological RAG                                                                                            | Must be checked against latest report and slides                                | Final target-author count, data volume, compute, exact base model | Treat as current provisional truth until local audit completes                     |

## Canonical terminology

Several architecture terms should be tightened immediately.

**PersonaRAG** should mean **retrieval conditioned on target-persona information**, not simply “RAG over documents written by the author.” The published PersonaRAG architecture analyzes user profile/context/live interaction and incorporates user-centric agents into retrieval and ranking. <sup>10</sup> Your implementation can

adapt that principle for authors, but it should be documented as **an author-persona adaptation inspired by PersonaRAG**, unless it actually reproduces the paper's multi-agent design.

**Ontology-RAG** should mean **RAG whose indexing/retrieval is explicitly constrained or enriched by a declared ontology/schema**. A graph automatically extracted from text is not sufficient by itself to justify the “ontology” label. OG-RAG explicitly separates the ontology from the document-derived structured representation. <sup>11</sup>

**LightRAG/GraphRAG** should be called graph-based retrieval methods. LightRAG indexes entities and relations and uses low-/high-level retrieval; Microsoft's GraphRAG constructs an entity knowledge graph and community summaries for local/global sensemaking. <sup>15</sup> Either could help implement a graph layer, but neither should be relabeled “ontology RAG” unless a project-defined ontology is imposed.

**QLoRA** is a training mechanism, not a persona architecture. It efficiently parameterizes adaptation through low-rank modules over a quantized base model. <sup>16</sup> The persona architecture is therefore more accurately named “**persona SFT using QLoRA**.”

**CharLoRA** is not interchangeable with standard LoRA/QLoRA. CharacterBot's CharLoRA introduces shared persona-specific structure and task-specific adaptation within a multi-task persona framework. <sup>3</sup>

**APR** in the CharacterBot paper means **Authorial Perspective Reframing**: converting implicit authorial perspective into explicitly attributed formulations intended to strengthen association between views and the author. <sup>3</sup> It is a data transformation/training strategy, not a standalone inference architecture.

**RAFT** is retrieval-augmented fine-tuning intended for a known-domain open-book environment, including training with useful and distracting documents. <sup>9</sup> It should not be called a generic superior version of RAG or QLoRA.

**Canonical unresolved parameters**

These should remain explicitly marked rather than guessed in final documentation:

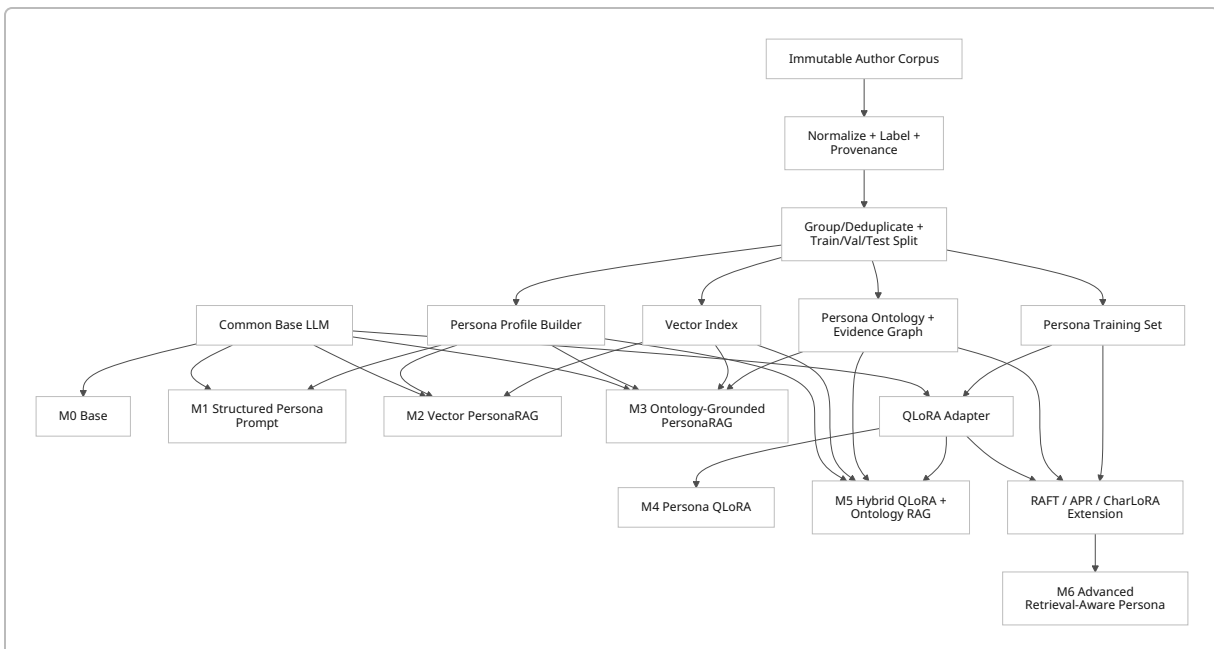
| Parameter              | Current state                                      | Proposed default until verified                                                                                |
|------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Exact target authors   | <b>Unspecified in accessible context</b>           | Preserve author ID as a first-class field                                                                      |
| Corpus size per author | <b>Unspecified</b>                                 | Report documents, tokens, platform and date coverage per author                                                |
| Available GPU/VRAM     | <b>Unspecified</b>                                 | Design for a single 24-GB-class GPU where practical; benchmark actual memory before freezing training settings |
| Canonical base model   | <b>Not verified against latest local artifacts</b> | <code>meta-llama/Llama-3.1-8B-Instruct</code> for controlled M0–M6 comparisons if licensing/hardware permit    |

| Parameter                 | Current state       | Proposed default until verified                                                                      |
|---------------------------|---------------------|------------------------------------------------------------------------------------------------------|
| Dataset split             | <b>Not verified</b> | Group-aware 80/10/10 default after dedup/cross-post clustering                                       |
| Embedding model           | <b>Unspecified</b>  | Freeze one embedding model for all retrieval comparisons                                             |
| Vector store              | <b>Unspecified</b>  | FAISS/Chroma-class local vector store; implementation choice should not become the research variable |
| Ontology storage          | <b>Unspecified</b>  | JSON/NetworkX first; graph database only if required by scale                                        |
| Primary evaluation metric | <b>Unspecified</b>  | Multi-axis evaluation rather than one score                                                          |

Meta's Llama 3.1 8B Instruct remains a defensible *experimental-control* default because it is an 8B instruction-tuned model with a large context window and is already present as a baseline in directly relevant persona research; this recommendation is about comparability and feasibility, not an assertion that it is the newest or strongest model available in 2026. <sup>17</sup>

## Architecture and experimental design

### Recommended architecture relationship



## Experimental ladder

**M0 — Base LLM.** No target-specific profile, retrieval, or fine-tuning. This establishes the essential control: how much apparent author similarity comes from general model knowledge rather than the project pipeline.

**M1 — Structured persona prompting.** The prompt contains a compact, frozen persona profile derived only from permitted training/profile data. LaMP found profile augmentation useful across personalized tasks, making this an important low-cost baseline before more complex retrieval or training. <sup>8</sup>

**M2 — Vector PersonaRAG.** Retrieve author-authentic passages using semantic/lexical relevance and attach them with provenance. This tests non-parametric personalization. Classical RAG separates the model's parametric memory from external retrieved evidence, while personalization work shows profile/history retrieval can improve personalized generation. <sup>18</sup>

**M3 — Ontology-grounded PersonaRAG.** The same base generator and approximately the same evidence corpus should be used as M2, but retrieval/reranking additionally uses the persona ontology. This is the clean experiment for your professor's ontological-RAG recommendation. OG-RAG's central idea is that structured domain concepts and relations can guide context selection beyond unstructured semantic similarity. <sup>11</sup>

Recommended persona ontology:

```
Author
├─ authored → SourcePost
├─ has_style_trait → StyleTrait
└─ expresses → Stance

SourcePost
├─ discusses → Topic
├─ mentions → Entity
├─ references → Event
├─ has_content_type → ContentType
├─ contains_evidence_for → Stance
├─ repost_of → SourcePost
└─ created_at → Time

Stance
├─ about → Topic
├─ polarity → support/oppose/mixed/uncertain
└─ supported_by → SourcePost

ContentType
├─ authentic_discourse
├─ author_promotion
└─ templated_promotion
```

- └ repost
- └ quote
- └ platform\_boilerplate
- └ uncertain

A query such as “How would this author discuss regulation of AI?” would first extract concepts such as `Topic=AI`, `Topic=regulation`, and perhaps relevant entities. Retrieval would then combine vector candidates with ontology-neighborhood evidence, prioritize primary author-authored material expressing stances on connected topics, apply promotional/noise rules, and return passages with source IDs. That is an adaptation of ontology-grounded retrieval to persona research; it is an inference from OG-RAG’s domain-oriented design rather than something OG-RAG itself tested. <sup>11</sup>

**M4 — Persona QLoRA.** Fine-tune a common base model using curated author-authentic training examples. QLoRA provides the efficient adaptation mechanism while leaving most base-model parameters frozen/quantized. <sup>16</sup> The research question becomes whether persona signals internalized parametrically outperform prompting/retrieval alone.

**M5 — Hybrid QLoRA + Ontology PersonaRAG.** Load the persona adapter and supply structured retrieved evidence. This asks whether **style/perspective internalization** and **explicit evidence access** are complementary. This is likely the strongest scientifically meaningful end-state architecture because it cleanly combines the two major adaptation families rather than simply adding more unrelated techniques.

**M6 — Advanced retrieval-aware persona model.** Train with retrieved relevant and distracting evidence in a RAFT-like setup. RAFT explicitly trains a model to operate in an imperfect retrieval environment, distinguishing useful from distractor documents. <sup>9</sup> APR and/or CharLoRA can be included here as optional ablations if schedule and compute permit; CharacterBot demonstrates that APR plus task-structured LoRA can improve both linguistic and deeper-perspective modeling in its author case study. <sup>3</sup>

## Variant reconciliation

| Variant           | Core or extension | Required data                             | Compute | Primary metrics                                | Research question answered              |
|-------------------|-------------------|-------------------------------------------|---------|------------------------------------------------|-----------------------------------------|
| Structured prompt | M1 core           | Persona summary/<br>profile               | Low     | style, stance,<br>task quality                 | Is explicit profile information enough? |
| PersonaRAG        | M2 core           | Author corpus +<br>profile +<br>retriever | Medium  | Recall@k,<br>persona fidelity,<br>evidence use | Does author-specific retrieval help?    |

| Variant      | Core or extension            | Required data                                        | Compute                                        | Primary metrics                                      | Research question answered                                            |
|--------------|------------------------------|------------------------------------------------------|------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------|
| Ontology-RAG | M3 core                      | Corpus + ontology + entity/ topic/ stance extraction | High implementation effort; moderate inference | retrieval recall/nDCG, grounding, stance consistency | Does explicit relational structure improve over flat retrieval?       |
| QLoRA        | M4 core                      | Clean persona SFT examples                           | High training effort                           | style, content/ stance, generalization               | Can persona be learned parametrically?                                |
| Hybrid RAG   | M5 core                      | QLoRA adapter + retrieval corpus                     | Med-High                                       | same metrics plus latency/ cost                      | Are training and retrieval complementary?                             |
| RAFT         | M6 optional/ advanced        | QA/ instruction examples + relevant/ distractor docs | High                                           | retrieval robustness, factuality, persona fidelity   | Can the trained model learn to use imperfect RAG better?              |
| APR          | M4/M6 optional preprocessing | Author-authentic text + reframing pipeline           | Med-High                                       | stance/ perspective understanding                    | Does explicit author attribution improve viewpoint learning?          |
| CharLoRA     | M4/M6 advanced alternative   | Multi-task persona datasets                          | High                                           | style + content/ perspective + task metrics          | Does structured multi-task persona PEFT outperform ordinary QLoRA?    |
| Persona-Plug | Comparator/ extension        | Complete author history + learned history encoder    | Med-High implementation                        | personalized generation metrics                      | Can a holistic history embedding replace retrieval/per-author tuning? |

| Variant  | Core or extension                   | Required data                      | Compute       | Primary metrics                           | Research question answered                                           |
|----------|-------------------------------------|------------------------------------|---------------|-------------------------------------------|----------------------------------------------------------------------|
| LightRAG | Retrieval implementation/comparator | Graph extraction corpus            | Medium        | retrieval quality, response quality, cost | Does automatically constructed graph retrieval help?                 |
| GraphRAG | Optional corpus-level comparator    | entity graph + community summaries | High indexing | global/thematic answer quality            | Does corpus-level graph summarization help global persona questions? |

Persona-Plug's defining distinction is that it represents a user's complete history through a lightweight embedder rather than per-user base-model training, while PersonaRAG centers on adaptive personalized retrieval; these therefore should not be collapsed into one “personalized RAG” category. <sup>19</sup> Likewise, RAFT specifically trains for an open-book domain setting with distractor documents, so it should only be added if retrieval-use robustness itself is a research question. <sup>9</sup>

## Evaluation contract

Every M0–M6 experiment should run on the **same frozen evaluation prompts** and emit a structured record containing:

```

experiment_id:
model_id:
base_model:
adapter_id:
retrieval_mode:
ontology_version:
prompt_version:
dataset_version:
test_split_hash:
generation_config:
retrieved_source_ids:
output:
latency:
metrics:
```

The primary evaluation should be multidimensional:

| Axis                         | Recommended measurement                                                |
|------------------------------|------------------------------------------------------------------------|
| Persona/style fidelity       | blinded human rating + style classifier/embedding/features             |
| Perspective/stance fidelity  | evidence-referenced rubric rating against held-out author evidence     |
| Content usefulness           | human or rubric-based quality score                                    |
| Retrieval                    | Recall@k, MRR/nDCG, evidence precision                                 |
| Groundedness                 | fraction of claims supported by retrieved/author evidence              |
| Hallucination/misattribution | unsupported author-position rate                                       |
| Memorization                 | n-gram/near-duplicate similarity to training/retrieval text            |
| General capability           | small non-persona sanity set to detect severe adaptation degradation   |
| Efficiency                   | latency, prompt tokens, index cost, training time/VRAM where available |

CharacterBot is a useful precedent for **separating style from content/perspective**, rather than pretending one lexical-overlap metric represents persona fidelity: the study reports task accuracy, content scores, style scores, BLEU/ROUGE/style matching and human assessment. <sup>3</sup> LaMP likewise treats personalization as a distinct evaluation dimension across multiple generation/classification tasks. <sup>8</sup>

## Data contract and promotional-content cleaning

### Non-destructive data taxonomy

The cleaning pipeline should never treat “promotional” as synonymous with “garbage.” Instead, assign **content\_type**, **authorship\_status**, **quality flags**, and **confidence**.

| Category           | Definition                                                               | Persona profile                 | RAG index                        | QLoRA training                               | Evaluation             | Raw archive |
|--------------------|--------------------------------------------------------------------------|---------------------------------|----------------------------------|----------------------------------------------|------------------------|-------------|
| AUTHENTIC_AUTHOR   | Clearly author-generated substantive content                             | Include                         | Include                          | Include                                      | Eligible               | Preserve    |
| AUTHOR_PROMOTIONAL | Author-written promotion containing meaningful personal language/opinion | Down-weight/include selectively | Separate collection or down-rank | Usually exclude from core; optional ablation | Exclude from core eval | Preserve    |

| Category             | Definition                                              | Persona profile              | RAG index                                    | QLoRA training                      | Evaluation         | Raw archive      |
|----------------------|---------------------------------------------------------|------------------------------|----------------------------------------------|-------------------------------------|--------------------|------------------|
| TEMPLATED_PROMOTION  | Repetitive ads/CTA likely copied from campaign/template | Exclude                      | Quarantine                                   | Exclude                             | Exclude            | Preserve         |
| SPONSORED_DISCLOSURE | Sponsorship/affiliate material                          | Down-weight                  | Separate/down-rank                           | Exclude unless studying promo voice | Exclude            | Preserve         |
| PLATFORM_BOILERPLATE | Auto-generated platform language                        | Exclude                      | Exclude                                      | Exclude                             | Exclude            | Preserve         |
| REPOST_CROSSPOST     | Same/near-same content repeated across platforms/posts  | Representative only          | One canonical record + aliases               | One representative                  | Group before split | Preserve all raw |
| QUOTE_THIRD_PARTY    | Text primarily authored by someone else                 | Do not treat as author voice | Only as contextual evidence with attribution | Exclude from persona text           | Exclude            | Preserve         |
| INTERVIEWER_GUEST    | Transcript utterance not made by target author          | Exclude                      | Separate attribution                         | Exclude                             | Exclude            | Preserve         |
| LINK_ONLY_LOW_INFO   | Little linguistic signal beyond URL/title/hashtags      | Exclude                      | Usually exclude                              | Exclude                             | Exclude            | Preserve         |
| UNCERTAIN_AUTHORSHIP | Attribution ambiguous                                   | Quarantine                   | Quarantine or low-rank                       | Exclude                             | Exclude            | Preserve         |

| Category       | Definition                                   | Persona profile                       | RAG index                   | QLoRA training                      | Evaluation          | Raw archive                    |
|----------------|----------------------------------------------|---------------------------------------|-----------------------------|-------------------------------------|---------------------|--------------------------------|
| NEAR_DUPLICATE | High overlap with another record             | Representative only                   | Deduplicated                | Deduplicated                        | Cluster-aware split | Preserve                       |
| MIXED_CONTENT  | Authentic commentary plus promotion/template | Extract/label segments where possible | Authentic segment preferred | Authentic segment only after review | Usually exclude     | Preserve original + derivative |

The need for strict duplicate and split handling is not merely cosmetic. Research on LM corpora found that deduplication reduces memorized output and train-validation overlap, improving the integrity of evaluation. <sup>20</sup> This makes cross-post clustering particularly important for social-media corpora where the same message may appear on several platforms.

## Proposed record schema

```

record_id: "stable-id"
author_id: "target-author"
platform: "youtube|instagram|x|linkedin|podcast|other"
source_url_hash: "..."
published_at: "..."
scraped_at: "..."

raw_text: "immutable original"
normalized_text: "normalized derivative"

authorship_status:
  label: "target|third_party|mixed|uncertain"
  confidence: 0.0

content_type:
  primary: "authentic_author"
  secondary: []

quality_flags:
  - "none"

promo:
  score: 0
  reasons: []

duplicate:
  cluster_id: null

```

```

    canonical_record_id: null
    similarity: null

provenance:
    scraper_version: "..."
    cleaning_version: "..."
    ontology_version: "..."

usage:
    persona_profile: true
    rag: true
    qlora: true
    evaluation: false

split:
    group_id: "..."
    split: "train|validation|test"

```

## Heuristics and regex rules

Regex should generate **signals**, never deletion commands.

```

PROMO_PATTERNS = {
    "explicit_ad": r"\b(sponsored|paid partnership|advertisement|#ad|affiliate)
\b",
    "cta": r"\b(link in (?:my |the )?bio|sign up|register now|shop now|buy now|
pre[- ]?order|subscribe now)\b",
    "discount":
r"\b(?:save\s+\$\d+|\d{1,2}%\s*off|promo code|discount code|use code)\b",
    "scarcity": r"\b(limited time|while supplies last|ends (?:today|tonight|
soon)|last chance)\b",
    "event_sales": r"\b(tickets? (?:are )?(?:available|on sale)|reserve your
spot)\b",
}

```

Additional weak signals:

```

+2 explicit sponsorship disclosure
+2 clear sales/registration CTA
+1 discount/code language
+1 unusually high URL/affiliate-link density
+1 repeated template signature across many posts
+1 extreme hashtag/CTA burst
+1 near-duplicate of known campaign copy

```

A reasonable initial policy is:

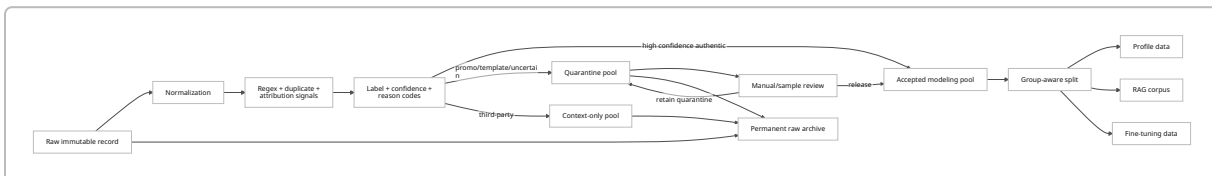
```
promo_score >= 4 → quarantine from persona training by default
promo_score 2-3 → mixed/promotional review queue
promo_score 0-1 → normal content path, subject to other quality rules
```

The threshold is a **project default to validate manually**, not a research result.

Template detection should go beyond regex:

1. lowercase + normalize URLs/usernames/numbers
2. remove platform formatting
3. compute exact normalized hash
4. compute MinHash/SimHash or embedding similarity
5. form duplicate/campaign clusters
6. preserve all raw records
7. choose one canonical representative for modeling
8. store cluster membership for split grouping

## Quarantine workflow



The essential principle is:

**Raw data is immutable. Cleaning produces annotations and derivative views. Nothing is silently destroyed.**

## Leakage policy

Create duplicate/cross-post clusters **before** assigning train/validation/test splits. Training examples and evaluation examples from the same near-duplicate cluster must never cross partitions. Deduplication literature shows why naive random splitting can inflate evaluation through overlap. <sup>20</sup>

For persona-generation evaluation, the safest default is:

```
TEST target output
  must not appear in:
    QLoRA training examples
    profile summaries derived from test material
```

vector index as an exact/near duplicate  
ontology evidence nodes derived from the same held-out output

For a separate explicitly labeled **open-book factual evaluation**, retrieval of relevant corpus evidence can be permitted by design, but that should be reported as a different evaluation regime. RAFT itself is specifically designed around such an open-book setting, reinforcing the need to distinguish it from closed-book generalization. <sup>9</sup>

## Canonical project documents and decision log

Draft `PROJECT_CONTEXT.md`

### # Project Context

#### ## Project

SJSU DATA298 capstone/research project on target-author/persona simulation using large language models.

#### ## Canonical Objective

Evaluate how progressively richer persona-adaptation mechanisms affect the ability of an LLM to generate responses aligned with a target author's linguistic style, recurring viewpoints, and documented evidence.

The project compares prompting, retrieval, ontology-grounded retrieval, parameter-efficient fine-tuning, and hybrid approaches under a common evaluation protocol.

#### ## Core Research Questions

- RQ1. How much persona fidelity is provided by a base LLM versus explicit persona prompting?
- RQ2. Does retrieval from authentic author content improve persona fidelity?
- RQ3. Does ontology-grounded retrieval improve over flat/vector PersonaRAG?
- RQ4. Does QLoRA-based persona adaptation outperform context-only approaches?
- RQ5. Are QLoRA and ontology-grounded retrieval complementary?
- RQ6. Optional: does retrieval-aware/structured persona training further improve fidelity and robustness?

#### ## Experimental Ladder

- M0 Base LLM
- M1 Structured Persona Prompt
- M2 Vector PersonaRAG
- M3 Ontology-Grounded PersonaRAG
- M4 Persona QLoRA
- M5 QLoRA + Ontology PersonaRAG
- M6 Advanced Retrieval-Aware Persona Model

## **## Data**

Primary data consists of public target-author content gathered from social platforms and/or related public sources.

Raw data is immutable.

Cleaning is annotation-based and non-destructive.

Promotional, templated, duplicate, third-party, and uncertain-attribution content is labeled and quarantined rather than silently deleted.

Exact author counts, corpus sizes and current dataset statistics:  
UNSPECIFIED UNTIL DATA INVENTORY IS VERIFIED.

## **## Evaluation**

Use identical frozen evaluation prompts across model variants.

Evaluate at minimum:

- persona/style fidelity
- perspective/stance fidelity
- content quality
- groundedness/misattribution
- retrieval quality for RAG models
- memorization/leakage
- efficiency

Do not treat BLEU/ROUGE alone as persona fidelity.

## **## Base Model**

Preferred experimental default:  
meta-llama/Llama-3.1-8B-Instruct

Status: PROPOSED, not frozen until repo/hardware/local-report audit confirms it.

## **## Ontological RAG**

Ontology-RAG is a required first-class experiment (M3), not merely future work.

Core ontology concepts:

Author, SourcePost, Topic, Stance, StyleTrait, Entity, Event,  
ContentType, Evidence, Time.

## **## Sources of Truth**

1. Course/professor requirements
2. Accepted DEC entries
3. PROJECT\_CONTEXT.md
4. ARCHITECTURE.md / DATA\_CONTRACT.md
5. Final report/current slides
6. Workbooks
7. exploratory notes

Conflicts must create a decision entry; do not silently choose one version.

### **## Current Critical Unknowns**

- Complete inventory of docs/research folder
- Exact latest report/slides architecture definitions
- Target-author count
- Dataset counts/tokens after cleaning
- Available GPU/VRAM
- Final base model
- Exact deadlines

## **Draft** ARCHITECTURE.md

### **# Architecture**

#### **## Principle**

Hold the base model, test set, generation settings, and evaluation protocol constant wherever possible. Each model should introduce one interpretable capability.

#### **## M0 – Base**

Base instruction model with no target-specific information.

Purpose:

Measure model prior / existing knowledge.

#### **## M1 – Structured Persona Prompt**

Base + frozen structured persona profile.

Purpose:

Measure prompt-only personalization.

#### **## M2 – Vector PersonaRAG**

M1 + retrieval over allowed target-author training corpus.

Retrieval:

hybrid lexical/vector or frozen vector retrieval.

Every returned chunk must retain source provenance.

Purpose:

Measure retrieval-based personalization.

#### **## M3 – Ontology-Grounded PersonaRAG**

M2 + explicit persona ontology and relation-aware retrieval/reranking.

Ontology:

Author, Source, Topic, Stance, StyleTrait, Entity, Event, ContentType, Evidence.

Purpose:

Test whether explicit relational persona structure improves retrieval, groundedness and perspective consistency.

### **## M4 – Persona QLoRA**

Common base + persona adapter trained on approved target-author data.

Purpose:

Measure parametric persona learning without retrieval.

### **## M5 – Hybrid**

M4 adapter + M3 retrieval.

Purpose:

Measure whether learned persona behavior and external structured evidence are complementary.

### **## M6 – Advanced Extension**

Retrieval-aware training using relevant and distractor context (RAFT-style), optionally incorporating APR and/or CharLoRA concepts.

Purpose:

Test whether the model can learn to use imperfect retrieval while preserving persona fidelity.

Status:

Optional until M0-M5 are reproducible.

### **## Non-Core Comparators**

Persona-Plug: low-parameter/history-embedding comparator.

LightRAG: graph retrieval implementation/comparator.

GraphRAG: global graph query comparator.

CharLoRA: advanced alternative to ordinary persona QLoRA.

These are not synonyms for core models.

## **Draft** DATA\_CONTRACT.md

### **# Data Contract**

#### **## Invariants**

1. Raw scraped data is immutable.

2. Every derived record links to its raw record.
3. Cleaning never silently deletes a source.
4. Every exclusion/quarantine action has a reason code.
5. Duplicate/cross-post grouping occurs before train/val/test assignment.
6. The frozen test set never enters QLoRA training.
7. Exact/near-duplicate target outputs never enter the RAG index used to evaluate those targets.
8. Ontology nodes retain links to source evidence.
9. Every dataset build has a version/hash.
10. Experiment results record the dataset version used.

## **## Content Labels**

AUTHENTIC\_AUTHOR  
AUTHOR\_PROMOTIONAL  
TEMPLATED\_PROMOTION  
SPONSORED\_DISCLOSURE  
PLATFORM\_BOILERPLATE  
REPOST\_CROSSPOST  
QUOTE\_THIRD\_PARTY  
INTERVIEWER\_GUEST  
LINK\_ONLY\_LOW\_INFO  
UNCERTAIN\_AUTHORSHIP  
NEAR\_DUPLICATE  
MIXED\_CONTENT

## **## Required Fields**

record\_id  
author\_id  
platform  
timestamps  
raw\_text  
normalized\_text  
authorship\_status  
content\_type  
quality\_flags  
promo\_score  
duplicate\_cluster  
provenance  
usage\_flags  
split

## **## Dataset Layers**

|                  |                                 |
|------------------|---------------------------------|
| data/raw/        | immutable                       |
| data/interim/    | normalized + labeled            |
| data/quarantine/ | reviewable derived manifests    |
| data/processed/  | approved modeling records       |
| data/splits/     | frozen train/val/test manifests |

Large/raw/private data should not be casually committed to Git.  
Store manifests, schemas, hashes, sample data and reproducible scripts in repo.

### **## Split Default**

Proposed starting split: 80/10/10 by duplicate/source grouping.

Do not freeze this percentage until corpus size per author is known.

## **Draft** DECISIONS.md

### **# Decision Log**

#### **## DEC-001 – Repository documents define persistent project context**

Status: ACCEPTED

PROJECT\_CONTEXT.md, ARCHITECTURE.md, DATA\_CONTRACT.md, DECISIONS.md, ROADMAP.md and SESSION\_HANDOFF.md form the durable context shared by ChatGPT, Cursor and the human researcher.

#### **## DEC-002 – Raw data is immutable**

Status: ACCEPTED

Cleaning creates labeled derivative datasets; records are not destructively deleted.

#### **## DEC-003 – Promotional content is labeled and quarantined**

Status: ACCEPTED

Promotion/template detection produces content labels, reason codes and confidence scores. Promotional data may be available for analysis but is not automatically used for persona fine-tuning or evaluation.

#### **## DEC-004 – Duplicate grouping precedes splitting**

Status: ACCEPTED

Exact duplicates, near duplicates and cross-posts must be clustered before train/validation/test assignment.

#### **## DEC-005 – Evaluation data is protected from leakage**

Status: ACCEPTED

Frozen evaluation target text cannot enter fine-tuning or persona profiles, and near-identical targets cannot be available to ordinary RAG evaluation.

#### **## DEC-006 – Persona retrieval is a core baseline**

Status: PROPOSED

M2 implements author-aware retrieval before graph/ontology complexity is introduced.

**## DEC-007 – QLoRA is the standard PEFT baseline**

Status: PROPOSED

Use a normal QLoRA persona model before implementing CharLoRA so that the effect of the advanced method remains measurable.

**## DEC-008 – Ontology-RAG is a first-class experiment**

Status: PROPOSED / PROFESSOR-GUIDED

M3 uses an explicit persona ontology and is directly compared with M2.

A generic automatically extracted graph is not by itself sufficient to call the experiment Ontology-RAG.

**## DEC-009 – APR is optional data augmentation, not a separate model**

Status: PROPOSED

APR may be tested as an M4/M6 training-data ablation.

**## DEC-010 – RAFT and CharLoRA belong after core experiments**

Status: PROPOSED

They add meaningful complexity and should not block completion of M0-M5.

**## DEC-011 – Persona evaluation is multi-dimensional**

Status: PROPOSED

Report style, perspective/content, grounding, retrieval, leakage and efficiency separately rather than collapsing the project into one metric.

**## DEC-012 – Persona-Plug is an extension/comparator**

Status: PROPOSED

PPlug answers an interesting low-parameter personalization question but is not required to establish the core prompting → retrieval → ontology → fine-tuning → hybrid ladder.

**## DEC-013 – Use one common base model where technically feasible**

Status: PROPOSED

Differences between architectures should not be confounded by changing the underlying LLM.

Current candidate: Meta Llama 3.1 8B Instruct.

Final status depends on hardware and existing project artifacts.

## **## DEC-014 – Final report numbers must be generated from experiment artifacts**

Status: PROPOSED

No manually maintained metric should exist only in slides/report.

Every reported result must trace to a stored run/metrics artifact.

### **Draft ROADMAP.md**

#### **# Roadmap**

##### **## Current Phase**

Context reconciliation + data-governance freeze.

##### **## Completion Order**

1. Enumerate research/project artifacts.
2. Reconcile workbook/report/slide contradictions.
3. Freeze DATA\_CONTRACT and dataset inventory.
4. Implement non-destructive cleaning/quarantine.
5. Create grouped frozen splits.
6. Build M0 and M1.
7. Build M2 vector PersonaRAG.
8. Build M3 Ontology-RAG.
9. Build M4 QLoRA.
10. Build M5 hybrid.
11. Evaluate M0-M5 under identical protocol.
12. Attempt M6 only if schedule permits.
13. Synchronize report, slides and demo with experiment artifacts.

##### **## Definition of Done**

An experiment is not complete until:

- config is saved
- dataset version is recorded
- random seed is recorded where relevant
- retrieval sources are logged
- outputs are stored
- metrics are computed
- failure cases are sampled
- report table can be regenerated

### **Draft SESSION\_HANDOFF.md**

#### **# Session Handoff**

## **## Current Canonical Direction**

The project evaluates target-author/persona simulation using a controlled M0-M6 architecture ladder.

Ontological RAG is M3 and is required as a direct comparison against vector PersonaRAG.

Promotional/noisy content is handled using labels and quarantine, not destructive deletion.

## **## Latest Accepted Decisions**

DEC-001 through DEC-005 are treated as working invariants.

DEC-006 through DEC-014 require confirmation against the latest local workbook/report/slides and implementation constraints.

## **## Current Blockers**

- ChatGPT has not yet enumerated the local docs/research folder.
- Exact current dataset statistics are unknown.
- Available GPU/VRAM is unknown.
- Exact base model in the latest report is unverified.
- Architecture/results in final report/current slides are unverified.

## **## Next Context Task**

Enumerate and ingest every file under:

C:\Users\okevi\SJSU\DATA298\298P\docs\research

Then:

- update RESEARCH\_INDEX
- create artifact comparison matrix
- mark conflicting claims
- supersede obsolete DEC entries
- promote context version from v0.9 to v1.0

## **## Next Implementation Task**

Do not implement advanced architectures until:

DATA\_CONTRACT, frozen splits and M0/M1 acceptance tests are committed.

## **## Handoff Rule**

At the end of every major ChatGPT/Cursor session update:

- current phase
- completed artifacts
- new decisions
- blockers
- next single implementation target
- files changed
- experiment IDs produced

This structure directly supports the SJSU expectation that the project demonstrate not only models but the complete data-analytics lifecycle and well-justified comparisons. <sup>21</sup>

## Prioritized implementation roadmap

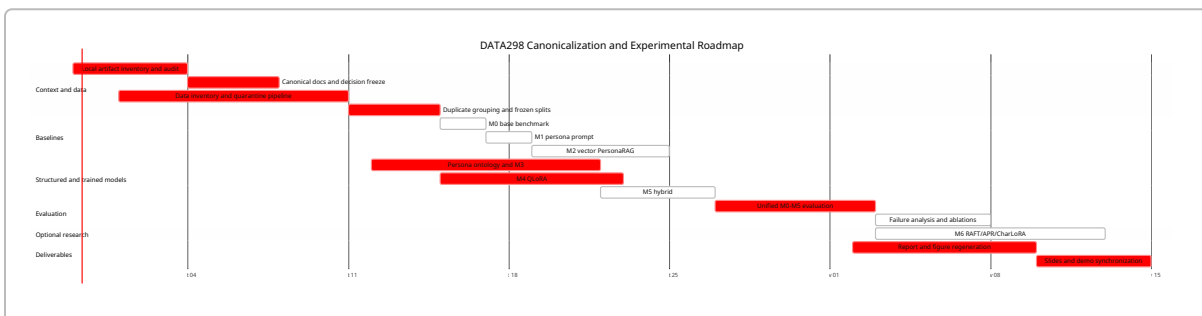
### Milestones

| Priority | Milestone                                                              | Effort   | Main blocker                      | Required Cursor/GitHub artifact                                 |
|----------|------------------------------------------------------------------------|----------|-----------------------------------|-----------------------------------------------------------------|
| P0       | Enumerate <code>docs/research</code> and complete local RESEARCH_INDEX | Medium   | Local filesystem not visible here | <code>docs/RESEARCH_INDEX.yaml</code> , inventory script/output |
| P0       | Reconcile workbooks/report/slides                                      | Medium   | Same                              | <code>docs/CONFLICT_AUDIT.md</code> , updated DEC entries       |
| P0       | Freeze data contract and raw inventory                                 | Medium   | Corpus stats                      | <code>DATA_CONTRACT.md</code> , <code>data_manifest.json</code> |
| P0       | Implement promo/attribution/duplicate labeling                         | Medium   | Threshold calibration             | cleaning modules + tests + quarantine manifest                  |
| P0       | Create leakage-safe grouped splits                                     | Medium   | duplicate clusters                | versioned split manifests                                       |
| P1       | M0 base benchmark                                                      | Low      | base model/environment            | run config + outputs + metrics                                  |
| P1       | M1 persona prompting                                                   | Low      | profile schema                    | frozen prompt/profile templates                                 |
| P1       | M2 vector PersonaRAG                                                   | Medium   | retrieval corpus                  | vector-index builder + retrieval eval                           |
| P1       | M3 ontology-grounded RAG                                               | High     | ontology schema/entity extraction | ontology schema + graph/index + retrieval tests                 |
| P1       | M4 QLoRA                                                               | High     | GPU + cleaned SFT corpus          | training config + adapter + checkpoints/metrics                 |
| P1       | M5 hybrid                                                              | Med-High | stable M3/M4                      | integrated inference config                                     |
| P1       | Cross-model evaluation                                                 | Medium   | all core outputs                  | unified results table + statistical analysis                    |

| Priority             | Milestone                       | Effort | Main blocker                     | Required Cursor/GitHub artifact         |
|----------------------|---------------------------------|--------|----------------------------------|-----------------------------------------|
| P2                   | M6 RAFT/APR/CharLoRA extension  | High   | time, compute, synthetic-data QA | optional experiment branch              |
| P2                   | Persona-Plug/LightRAG ablations | High   | time                             | optional comparator branches            |
| P0 before submission | Report/demo synchronization     | Medium | reproducible results             | generated figures/tables + final slides |

## Default milestone timeline

The dates below are a planning default beginning immediately after September 28, 2026; they are not inferred course deadlines.



The key scheduling principle is to **delay M6 until M0–M5 are reproducible**. RAFT and CharacterBot-style CharLoRA/APR are legitimate research extensions, but both introduce additional training-data construction and training methodology that could otherwise consume time without answering the basic controlled comparisons first. RAFT requires relevant/distractor training contexts, while CharacterBot generates multiple persona-specific tasks and APR-transformed training material before its task-specific CharLoRA stage. <sup>22</sup>

## Cursor/GitHub acceptance gates

Cursor should not be told simply to “implement Ontology-RAG.” It should receive explicit acceptance criteria such as:

### M3 acceptance criteria

- [ ] Uses exactly the M2 generator/base-model family.
- [ ] Uses the same eligible RAG evidence corpus as M2.
- [ ] Loads a versioned ontology schema.
- [ ] Maps evidence records to ontology concepts/relations.
- [ ] Logs matched ontology concepts for every query.
- [ ] Logs retrieved source record IDs.

- [ ] Never indexes quarantined records unless experiment config explicitly allows it.
- [ ] Passes evaluation-leakage tests.
- [ ] Evaluates the same frozen prompts as M0-M2.
- [ ] Can disable ontology-aware reranking to reproduce an M2-like control.

That makes M3 a scientific ablation instead of merely a different software stack.

Likewise, M4 should not proceed until the training manifest is frozen:

#### M4 acceptance criteria

- [ ] Training records come only from approved training partition.
- [ ] No frozen-test duplicate cluster appears in training.
- [ ] Promotional/template exclusions are recorded.
- [ ] Base model and tokenizer are pinned.
- [ ] QLoRA configuration is version-controlled.
- [ ] Adapter artifacts have run IDs.
- [ ] Evaluation uses the exact same test prompts as M0-M3.

QLoRA is specifically valuable here because it gives a strong parameter-efficient fine-tuning baseline without requiring full-model adaptation. <sup>16</sup>

## Parallel-work boundaries for this chat and the GitHub/Cursor chat

The most efficient division is to keep **research truth and scientific decisions here**, while keeping **repository state and execution there**.

| This project-context / deep-thinking chat             | GitHub / Cursor implementation chat              |
|-------------------------------------------------------|--------------------------------------------------|
| Audit research evidence                               | Enumerate actual repo/files                      |
| Reconcile contradictory architecture claims           | Create/update Markdown/YAML files                |
| Decide what M0-M6 scientifically mean                 | Implement M0-M6 components                       |
| Define ontology concepts/relations                    | Implement graph/index/storage                    |
| Decide data-use policy for promotional/noisy material | Write cleaners, classifiers, manifests and tests |
| Define leakage rules                                  | Implement duplicate detection/split validation   |
| Review whether an experiment answers its claimed RQ   | Run training/inference                           |
| Interpret results and failure cases                   | Save metrics/output artifacts                    |

| This project-context / deep-thinking chat       | GitHub / Cursor implementation chat    |
|-------------------------------------------------|----------------------------------------|
| Accept/supersede DEC entries                    | Commit accepted decisions              |
| Decide which advanced papers justify extensions | Implement only approved extensions     |
| Validate report claims against research         | Regenerate tables/figures from results |
| Maintain canonical conceptual path              | Maintain canonical executable state    |

A practical handoff cycle is:

```

Research/context chat
↓
"DEC-008 accepted: M3 must use explicit persona ontology"
↓
DECISIONS.md / ARCHITECTURE.md specification
↓
Cursor implements + tests
↓
experiment artifacts/results
↓
Research/context chat interprets evidence
↓
decision retained, modified, or superseded

```

The repository should therefore become the **persistent shared memory**, while chats act as reasoning and execution interfaces around it. The canonical files prevent Cursor from inventing an architecture from whichever source file it happens to inspect, and they prevent a new ChatGPT planning session from reconstructing the project differently each time.

The immediate context state is best labeled:

**Canonical Project Context v0.9 — architecture/data-policy scaffold validated against primary research and official SJSU program requirements; local research-folder enumeration and workbook/final-report/slide reconciliation still pending.**

That qualification matters. The externally verified evidence already strongly supports the experimental sequence: classical RAG provides the retrieval baseline; personalized retrieval literature supports profile-aware context; explicit ontology grounding creates a scientifically distinct structured-retrieval treatment; QLoRA provides the efficient parametric-adaptation baseline; CharacterBot supplies directly relevant evidence for APR/CharLoRA-style author simulation; and RAFT provides a defensible advanced experiment for learned use of imperfect retrieval. <sup>23</sup> The missing piece is not conceptual architecture research—it is **file-by-file reconciliation of your actual local project artifacts**, which cannot be truthfully completed until that Windows directory is available to the conversation.

- 1 21 [https://www.sjsu.edu/applied-data-science/docs/masdi\\_course\\_assessment\\_table.pdf](https://www.sjsu.edu/applied-data-science/docs/masdi_course_assessment_table.pdf)  
[https://www.sjsu.edu/applied-data-science/docs/masdi\\_course\\_assessment\\_table.pdf](https://www.sjsu.edu/applied-data-science/docs/masdi_course_assessment_table.pdf)
- 2 11 <https://arxiv.org/html/2412.15235>  
<https://arxiv.org/html/2412.15235>
- 3 <https://arxiv.org/html/2502.12988v3>  
<https://arxiv.org/html/2502.12988v3>
- 4 19 <https://arxiv.org/abs/2409.11901>  
<https://arxiv.org/abs/2409.11901>
- 5 14 18 23 <https://arxiv.org/html/2005.11401v1>  
<https://arxiv.org/html/2005.11401v1>
- 6 <https://arxiv.org/html/2106.09685>  
<https://arxiv.org/html/2106.09685>
- 7 16 <https://arxiv.org/html/2305.14314v1>  
<https://arxiv.org/html/2305.14314v1>
- 8 <https://arxiv.org/html/2304.11406v1>  
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- 9 22 <https://arxiv.org/html/2403.10131>  
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- 10 <https://arxiv.org/html/2407.09394v2>  
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- 13 <https://www.microsoft.com/en-us/research/publication/from-local-to-global-a-graph-rag-approach-to-query-focused-summarization/>  
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<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>
- 20 <https://arxiv.org/html/2107.06499v2>  
<https://arxiv.org/html/2107.06499v2>